

SCC.213 Internet Applications – Coursework Task

Use of Generative AI in SCC.213

For this module, Generative AI is **not allowed**. The reason is that it can limit your grasp of key concepts and encourage reliance on tools instead of building important problem-solving skills — especially when you're working with technologies you're not experienced with. By excluding AI, you'll have the chance to deepen your understanding and be better prepared for the industry.

The university's full position on AI in teaching and learning is available [here](#).

Please note: We expect you to use the techniques, technologies and libraries that we cover on the SCC.213 module. If your coursework submission uses techniques, technologies and libraries not covered on the module your work will be scrutinised for potential plagiarism and, at best, an appropriate penalty applied to your grade.

Coursework: Connect4 Game

The aim of this game is to strengthen your web development skills while building a fun, interactive, and fully functional game. You'll design and implement a Connect4 game (https://en.wikipedia.org/wiki/Connect_Four) based on the work introduced in the weekly lab worksheets. You'll implement a fully integrated client-server game where players can connect, play in real-time, and store scores in a database.

Connect4 is typically played with a 7x6 vertical 'board' where two players take turns to drop different coloured discs into the columns, with the winner being the first to create a line of four of their coloured discs (horizontally, vertically, or diagonally). If every position on the board is filled without a winner this results in a draw. Despite its simplicity, the game requires strategic thinking and planning, making it a great exercise for learning core web development concepts.

This game has several key differences to the noughts and crosses (tic-tac-toe) game used as an example in the labs:

- **The Board:** Connect4 typically uses 7 columns and 6 rows.
- **Placing Pieces:** The playing pieces are physically dropped into the columns and so must stack from bottom to top (i.e. cannot be placed anywhere)
- **Additional complexity:** With a 7x6 board there are around 2 trillion potential winning combinations so hard-coding the different winning possibilities and using lookup is impractical.

Variations

The game will support the following modes:

1. **Player vs. Player** : Two players take turns playing in the same web browser.
2. **Player vs. AI**: Play against the computer with different levels of difficulty:
 - **Random AI**: Makes random moves.

- **Intermediate AI:** Attempts to win or block the player strategically.
 - **Advanced AI:** Uses algorithms like Minimax for optimal moves.
3. **Network Player vs. Player:** Two players take turns playing over the network/different web browsers on the same machine.

Coursework Objectives

The primary goal is to build a web application that implements Connect4 with a modern design and interactive gameplay. The project will help you learn about front-end and back-end development, client-server interactions, and database integration.

Key Tasks:

1. **Design the User Interface:** Build a simple, intuitive game board using HTML and CSS. The user interface should be responsive across mobile, tablet, and desktop. It should handle player moves by updating the board appropriately when a disc is 'dropped' into a column and preventing moves in a full column.
2. **Integrate a Server:** Add a backend using Node.js and Express.js to handle game logic and allow two players to connect and play remotely. It should also allow a player to play against the AI. Note: AI should be implemented at the backend (not the front end/browser).
3. **Implement Core Game Logic:** Use JavaScript to handle game rules and manage the player vs. player mode. If all spaces on the board are filled it should declare that the game is over. A reset function should allow players to restart the game, clearing the board and ensuring turns reset correctly.
4. **Add a Database:** Maintains a database table holding player name (/alias) and number of games started. Database table should be stored at the server side, content from the database should be shown on the main game page in a coherent way.
5. **Winning:** The game should check for a winner by detecting four matching symbols in a row, column, or diagonal, highlighting the winning combination and displaying a message. Number of wins should be integrated with the database to make a high scores table).
6. **Add AI Functionality:** Extend the game with an AI opponent, offering multiple difficulty levels. The player should be able to choose between Player vs. Player or Player vs. AI mode. If Player vs. AI is selected, allow the player to choose a difficulty level: naïve (random moves), intermediate (blocking and winning strategies), or advanced (using the minimax algorithm). Based on the selection, the game should either wait for another human player's input or generate AI moves accordingly. Ensure that the AI moves are executed correctly within the game loop, maintaining turn order and game rules. This should be implemented at the server side (Node.js).

7. **Final Integration and Deployment:** Combine all components into a polished application (integrating all the aspects above) and test thoroughly.

Accessibility requirements

Ensure the user interface follows accessible web design principles, making the game playable for all users, including those with disabilities. Use proper HTML semantics, provide keyboard navigation support, ensure sufficient colour contrast, and include ARIA attributes where necessary. The game should also be responsive, adapting smoothly to mobile, tablet, and desktop screens while maintaining usability.

Submission guidelines

The assessment of this coursework includes **three** important parts which must all be complete by the deadline. This includes:

- 1) An in-person code Q&A session with a member of teaching staff in your allocated lab session¹. **These will take place in the weeks 19 and 20 scheduled lab sessions.** This will take around 5 minutes and involve answering technical questions on your coursework implementation.
- 2) Submission of a video demo/code walkthrough (screen record with voice-over²) of around 5 minutes (8 minutes max), on Moodle, which shows all aspects of the marking scheme you have implemented in your user interface along with brief explanations of how you have implemented them in your code. Try to focus on the more interesting and challenging aspects of your solution and clearly explain how you decided to implement them.
- 3) Submission of your complete project source on Moodle as a ZIP file, including all source files you created (HTML, CSS, and JavaScript), **please do not include node libraries in your zip file**. Ensure all source code is well-structured and documented. Additionally, complete the feature implementation form (to be published near the submission deadline) to indicate which features have been successfully implemented and any extra enhancements made.

The deadline for submission is Friday, Week 20, at 4 PM. In order to achieve a pass mark you must complete all three parts of the assessment by the deadline.

Marking scheme

The coursework comprises 20% of the overall module mark and will be marked out of 20, the breakdown of marks across the three aspects of the task is specified below. Your overall mark will be aggregated (i.e. you do not need to gain all marks in aspect 1 in order to gain marks in aspect 2 or 3, but failing to do so may limit your mark).

¹ If you feel this will be challenging for you and have an appropriate ILSP in place please contact module staff in advance of the deadline so appropriate arrangements can be made.

² You may wish to use Microsoft Steam Screen Recorder - <https://support.microsoft.com/en-gb/office/microsoft-stream-screen-recorder-e98d8791-2b82-4dc7-889a-959724e3cbad> - this only works in Google Chrome and Microsoft Edge web browsers and has a maximum recording length of 15 minutes.

Marks will only be awarded where a student has been able to demonstrate adequate understanding of technical aspects in the Q&A session, and adequately explained aspects of their implementation in the video demo/code walkthrough. Adequate explanation would typically show understanding of key technical aspects and include specifics of how you applied them in your code. Inadequate explanation would typically be inaccurate, vague, unspecific or refer to technical aspects in generalities rather than in relation to your code.

1. Functionality & Game Logic (9 Marks)

- **Basic Gameplay (3 Marks):** Game correctly implements player moves (up to 3 marks depending on games modes implemented), systems checks for a complete board and resets properly (1 mark).
- **Winning (3 Mark):** Game can reliably detect a winner in all game modes in a 7x6 board and react appropriately.
- **AI Implementation (3 Marks):** AI supports naïve (1 Mark), intermediate (1 Mark), and advanced (1 Mark) difficulty levels.

2. User Interface & Design (4 Marks)

- **Responsive Design (2 Marks):** UI adapts seamlessly from desktop to mobile and tablet (1 mark for each, 0 marks if no responsive design implemented).
- **Accessibility Features (2 Marks):** Game follows WCAG and WAI-ARIA accessibility guidelines: e.g., contrast, keyboard navigation, ARIA attributes - approx. 0.5 mark per correctly implemented guideline within implementation depending on complexity.

3. Server-Side & Database Integration (7 Marks)

- **Client-Server Communication (4 Marks):** Server correctly processes and propagates game state to network player vs. player - makes effective use of AJAX or WebSockets (up to 2 marks for only AJAX, up to 3 marks for WebSockets with or without AJAX), all data encoded in JSON (1 mark).
- **Database Integration (3 Marks):** Player scores are stored and retrieved correctly from the database: SQLite database integrated into Node.JS implementation (1 mark), SELECT/INSERT implemented appropriately in JavaScript (1 mark), appropriate scoring mechanism implemented e.g. top high scores visible on game UI (1).

Marks will be given based in the completeness, robustness and quality of the implementation. You are expected to use the technologies and techniques taught on the module; you may be penalised if you do not.